

Enterprise Retail Intelligence & Customer Analytics Platform

1. Project Overview

End-to-end Retail Analytics project built using Python, SQLite SQL, and Power BI.

2. Business Problem

Analyze customer purchasing behavior and provide business insights for better decision-making.

3. Business Objectives

Analyze sales, customers, products, and regional trends through interactive dashboards.

4. Dataset Overview

Dataset: shopping_trends_updated.csv

Rows: 3900

Columns: 18

Missing Values: 0

Duplicate Rows: 0

5. Tools & Technologies

Python, Pandas, NumPy, SQLite, SQL, Google Colab, Power BI.

6. Analytics Workflow

Business Understanding → Data Inspection → SQL Analysis → EDA → Dashboard Development → Insights.

7. KPIs

Total Revenue, Total Customers, Average Purchase, Average Rating, Previous Purchases.

8. Dashboard Pages

Executive Overview

Customer Analytics

Product Analytics

9. Key Insights

Clothing generated the highest revenue. Customer behavior varies by age, subscription status, payment method, and season.

10. Recommendations

Improve customer retention, focus on top-selling products, optimize seasonal campaigns, and target high-performing regions.

11. Conclusion

The project demonstrates practical business analytics skills using Python, SQL, and Power BI.